

Accelerometer Sensor

Objective:

To read and analyze acceleration values along the X, Y, and Z axes using the built-in LSM9DS1 accelerometer sensor on the Arduino Nano 33 BLE.

Note: Install Arduino_LSM9DS1 Library

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_LSM9DS1 library.
3. Opened the SimpleAccelerometer example.
4. File → Examples → Arduino_LSM9DS1 → SimpleAccelerometer
5. Uploaded the program to the Arduino Nano 33 BLE board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Observed the X, Y, and Z acceleration values while moving the board.

Working Principle:

The accelerometer measures acceleration along three axes:

- X-axis → Left and Right movement
- Y-axis → Forward and Backward movement
- Z-axis → Up and Down movement

The sensor measures acceleration in units of g, where: $1\text{ g} \approx 9.81\text{ m/s}^2$ When the board is stationary on a flat surface:

- $X \approx 0$
- $Y \approx 0$
- $Z \approx 1$

because gravity acts along the Z-axis.

Observations:

- X and Y values remained close to zero when the board was stationary.
- Z value remained close to 1 due to Earth's gravity.
- The values changed when the board was tilted or moved.

-0.26	-0.10	0.75
-0.27	-0.17	0.95
-0.29	-0.20	1.11
-0.29	-0.19	1.22
-0.28	-0.17	1.29